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Animal Detection and Classification in Image & Video Frames using YOLOv5 and YOLOv8

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***Abstract*—Wild animals often enter residential or agricultural areas, causing damage to property and having the potential to be dangerous to humans. This research study plan on developing a machine learning system that can automatically detect the presence of animals in surveillance footage and alert the appropriate authorities. Animal intrusion can lead to various problems such as including property damage, health risks, safety risks, and economic losses. Automatic animal detection has numerous applications in different fields including wildlife conservation. In this regard, two proposals utilizing the You Only Look Once (YOLO) object detection algorithm are presented. The first proposal focuses on animal detection in wildlife conservation, reporting high accuracy in identifying and classifying animals in video frames. The second proposal addresses the issue of animal intrusion into human-inhabited areas, using YOLO-based deep learning methods for object detection, tracking, segmentation, and edge detection. It also highlights the challenges of generalizing detection models from native habitats to deployment scenarios and suggests the production of synthetic data for training in a certain domain that is semi-automated. We worked on both versions YOLOv5 and YOLOv8 for detection and we received around 80% accuracy on YOLOv8. Comparatively YOLOv8 is better than YOLOv5 in terms of accuracy. These proposals showcase the effectiveness of using YOLO for animal detection and its potential to aid in animal population monitoring, behavior analysis, security, and wildlife conservation.**

***Index Terms*—YOLOv5,YOLOv8, Convolutional Neural Network.**

1. Introduction

Globally, nature is deteriorating at previously unheard-of speeds, and different human- caused changes to have speed up the loss of biodiversity. For ecologists and wildlife conservation scientists, camera trap surveys can offer useful data on the distribution of species diversity, animal behaviour, population density, community dynamics, and other topics. When animals pass by, a camera trap will automatically take pictures or videos. However, camera traps are also subject to complicated settings (such as plants moving with the wind, exposure to sunlight), which can lead to false triggers and provide pictures or films occasionally without any animals. It is necessary to clean and sort the gathered photographs and videos, which are very time- consuming and labour-intensive manual activities. Automated detection of animals in video frames is an important task in various fields, such as wildlife conservation, animal behaviour research, and veterinary medicine. This task is challenging due to the variability in animal appearance, shape, and size, as well as the complexity of the natural environments in which animals live.

To address this challenge, various computer vision algorithms have been developed, such as deep learning-based object detection systems. One of the most popular and effective deep learning- based object detection systems is You Only Look Once (YOLO). One image or video frame can contain several objects according to the real-time object detection method known as YOLO. In the context of animal detection in video frames, YOLO can be trained on annotated datasets of animal images or video frames. The trained model can then be used to detect and track animals in new video frames in real-time. This can provide valuable insights into animal behaviour, ecology, and physiology, as well as support conservation efforts by enabling the monitoring and protection of animal population.

2. LITERATURE SURVEY

This section discusses the work done by various authors, students, and researchers in the area of Animal Detection in Video Frames using Machine Learning as given in TABLE I. This section's goal is to provide a critical overview of what is currently known about animal detection.

TABLE I: Survey on Animal detection

Sl. No.	Abstract	Methodology	Conclusion
[1]	It is suggested to use CNN to classify the input animal photos.	Compare the suggested CNN approach to the overall recognition accuracy of the PCA, LDA, LBPH, and SVM. The database of wild animals is compiled for the experiments. Various numbers of training photos and test images were used to obtain the overall performances.	The experimental results demonstrate that the suggested strategy beats other tested methods and has a beneficial impact on overall animal recognition performance.
[2]	All visual media are interpreted by a computer as a collection of numerical values. They need image processing algorithms to inspect the contents of images as a result of this strategy.	The goal of this study is to determine which of three key image processing algorithms is the fastest and most effective. The effectiveness of these three algorithms is compared in this study utilising the Microsoft COCO dataset.	It was discovered that Yolo-v3 is the fastest, followed closely by SSD and Faster RCNN, who came in last.
[3]	The three most prevalent topologies and learning algorithms for CNN are provided in this paper along with a conceptual grasp of the network. This study will encourage scholars to pursue this area of research by giving them a thorough understanding of CNN.	Two crucial CNN learning algorithms. CNN has become a well-known method for classifying objects based on their surroundings. It has a phenomenal capacity for learning contextual cues and has so circumvented the difficulties associated with pixel-wise categorization.	In comparison to shallow networks, deep learning has the key advantage of being able to independently identify important features in high dimensional data.
[4]	The objective is to distract the animal without causing injury while also safeguarding the crops from animal-caused damage.	When an animal is present, the PIR and ultrasonic sensor alert the controller with an input signal. The APR board will turn on instantly, and a sound is played to distract the animal.	The primary goals are to safeguard crops from animal caused damage and to divert animals safely.
[5]	In this study, they looked at whether thermal imaging and digital image processing might be utilized to detect chickens and rabbits on their own in a grassland setting.	The measurement of the radiation mode of heat transport is the foundation of infrared thermography. On the basis of the video footage, they applied digital image processing algorithms to recognize animals.	Potential exists for the use of thermal imaging to identify animals, and detection rates were nearly 100%.
[6]	A focus on wildlife analysis and monitoring through the recognition of animals in natural scenes captured by camera-trap networks.	Created a model that, using the plan for the animal background verification model, checks the animal and backdrop patches from the camera-trap photos.	The results of the experiments demonstrate how effective and reliable the proposed method is for both daytime and nighttime wild animal detection.
[7]	Artificial intelligence (AI) systems may automatically learn from their experiences and get better over time thanks to a technique called machine learning. The main goal is to make it possible for computers to autonomously learn without aid from humans and modify their behaviour accordingly.	A particular kind of artificial neural network called a convolutional neural network (CNN) analyses data using the supervised learning method of the perceptron algorithm. CNNs are applicable to cognitive tasks such as natural language processing, image processing, and more.	This project uses (CNN) algorithm to detect wild animals. The algorithm classifies animals efficiently with good accuracy and also the image of the detected animal is displayed for a better result.

[8]	Considering how quickly India's population is growing, finding shelter for everyone is a significant concern. Because of this, the majority of people rely on the forest to meet their needs. Due to their proximity to the forest, people frequently encounter wild creatures, and in order to protect humans, these animals are frequently murdered. Now, it's crucial for both human and animal life to be protected.	Six steps make up the suggested methodology: Foreground Detector utilising Back Subtraction, Morphological Filter, Blob Analysis, Recognition using Backpropagation Algorithm, and Recognition of Result.	We can draw the conclusion that the background subtraction approach produces better and more accurate results for the detection of moving animals based on all of these findings and debates. Performance analysis displays the method's level of acceptance.
[9]	Nowadays, wildlife monitor has become challenging work, and that too without the help of technology. To remedy this, we came up with the solution of an animal classifier camera that will detect the animals in the forests and keep a record of the animals. The system is also useful in zoological parks to maintain wildlife.	The first step in the methodology is to make setup of Raspberry pi by installing Raspbian OS and interfacing of SD card with it. Installing the python idle in it as our system is totally based on python code and installing the required libraries like NumPy, pandas, and TensorFlow.	Due to the increased hunting of animals, it has become necessary to keep a record of wildlife animals and protect the environment. Also, it's difficult to implement the traditional ways of keeping the record.
[10]	Applications for automatic detection of stray animals in human-populated regions include crucial security and traffic safety measures. This study makes an effort to use deep learning to resolve this issue.	ImageNet was used for data collection. Different algorithms like YOLO, RCNN, and Deeplab were compared with each other.	Detection in images with unfamiliar backgrounds was difficult. Using segmentation masks helped the case.
[11]	Animal detection and classification let people distinguish between predators and nonpredators, both of which constitute a serious threat to people and the environment. Since it is frequently seen on roads and causes multiple collisions with cars, it lowers the rate of traffic accidents in diverse areas.	In order to create a multiscale attention mechanism that performs dot multiplication and SoftMax by multiscale features to get the weights, this article suggested using the ResNet-50 Network as a backbone. This mechanism would then be added to fine-tune the species' characteristics.	ResNet-50 Network has shown promising results with a multiscale attention mechanism and feature pyramid network which has adequately enhanced the detection of animal species.
[12]	Accidents involving wildlife and vehicles cause significant human, environmental, and financial losses, as well as harm to people and property. Wild and domestic animals that are on the road pose risks to other drivers. Most of the time, when drivers become aware of an animal in their route, it is too late to stop the car in time to prevent an accident. Therefore, the goal of this project is to create an animal detection system that alerts drivers to potential animal collisions.	Ultrasound, Infrared, Radio Frequency and various other techniques.	Out of the hardware methods, available cameras were the most promising and feasible in the long run. It overcame most of the disadvantages of other technologies.

3. PROBLEM DEFINITION

Wild animals often enter residential or agricultural areas, causing damage to property and having the potential to be dangerous to humans. We plan on developing a machine learning system that can automatically detect the presence of animals in surveillance footage and alert the appropriate authorities. The system will aim at being able to accurately identify a range of different animals including (but not limited to) bears, tigers and elephants. The system should be able to process the given image or video with minimal latency and false positives.

4. METHODOLOGY

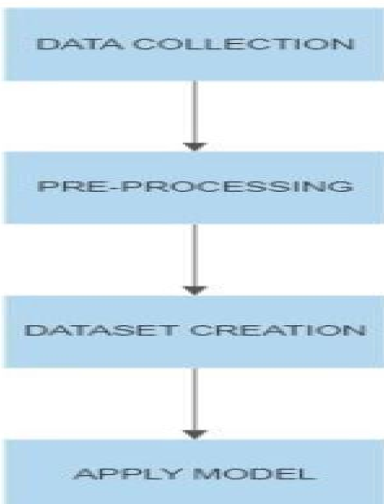

Fig. 1. System design

The suggested system architecture for the animal detection system is shown in Fig. 1. We present a detailed analysis of our implementation of two object detection models, YOLOv5 and YOLOv8, for the specific problem definition provided. We have created a user interface for our project. We can provide images or videos as input data to be processed with also the option of live input. Later the backend processes the input using the trained YOLOv5 or YOLOv8 model (based on different datasets).

A. YOLOv5

A deep learning architecture called YOLOv5 was developed for real-time object detection tasks. The YOLO (You Only Look Once) models are renowned for their quick and accurate object detection capabilities in pictures and movies.

B. YOLOv8

The most recent iteration of the YOLO algorithm, known as YOLOv8, performs better than earlier iterations because to the addition of modules for spatial attention, feature fusion, and context aggregation.Because of these enhancements, YOLOv8 has become one of the most important object detection systems in the industry.

Requirements: Software Requirements: Python,Scikit Learn, Pandas, OpenCV, Pillow, PyTorch, Flask, YOLOv5, YOLOv8

Hardware requirements: 8-16gb GPU, 8-16gb CPU

C. Data Collection

• Dataset used is COCO dataset.

• Collection of photos of animals of various species.

• Data for the animals which is already available on the internet.

• The COCO dataset (short for Common Objects in Context) is a large-scale image recognition, segmentation, and captioning dataset.

• Kaggle offers public datasets, and it also provides tools for data exploration, visualization, and modelling, as well as a community forum for discussion and collaboration.

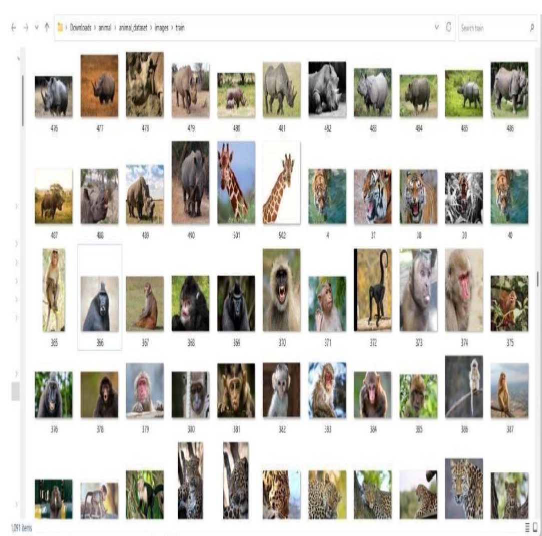

Fig. 2. Data Collection

D. Pre-processing

• We are using makesense.ai to convert the given set of images into data which can be fed to train the YOLO v5 algorithm. • It creates a zip file with all the details converted to the specified format.

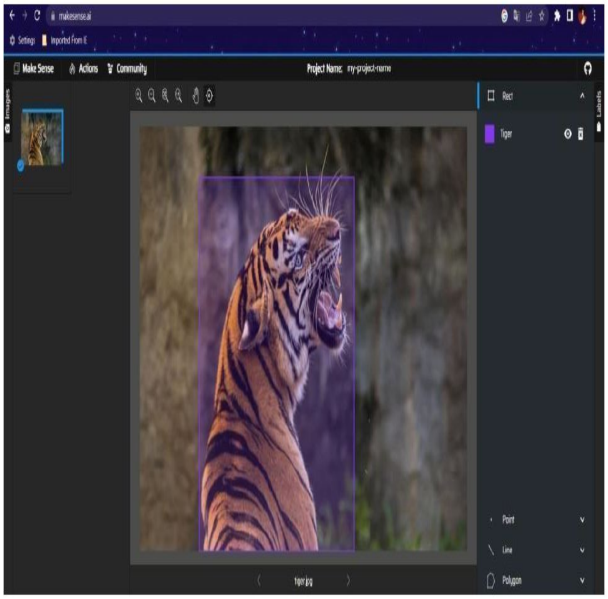

Fig. 3. Pre-processing

E. Dataset Creation

• The data from the different websites and the dataset from Kaggle and COCO are merged.

• The COCO dataset contains more than 330,000 images and over 2.5 million object instances labelled with bounding boxes, segmentations, and captions.

• The dataset is in the form of a text file for easy processing as it is shown in Fig.4.

• The dataset is split into 80:20 for training and testing.

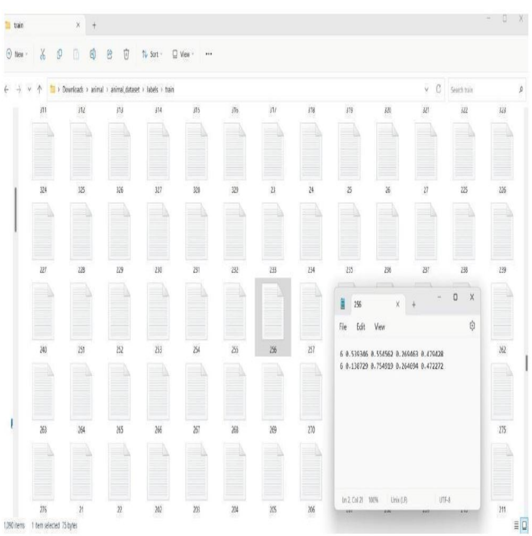

Fig. 4. Dataset Creation

F. Applying the Model

Yolov5 and Yolov8 are both versions of the popular object detection algorithm called YOLO (You Only Look Once). YOLO is a state-of-the-art deep learning model that can detect and classify objects in an image in real-time.

Applying YOLOv5 and YOLOv8:

1. Collect and prepare the dataset: Collect a dataset of images and annotations, and reprocess the images by resizing and normalizing them.

2. Install dependencies: Install the necessary dependencies for running YOLOv5 and YOLOv8, such as Python, PyTorch, and OpenCV.

3. Download pre-trained weights: Download pre-trained weights from the official GitHub repository.

4. Configure the model: Customize the configuration file to specify the input and output paths, image size, confidence threshold, and other parameters.

5. Use the model for object detection: Apply the model to the input data and visualize the results with bounding boxes, class labels, and confidence scores.

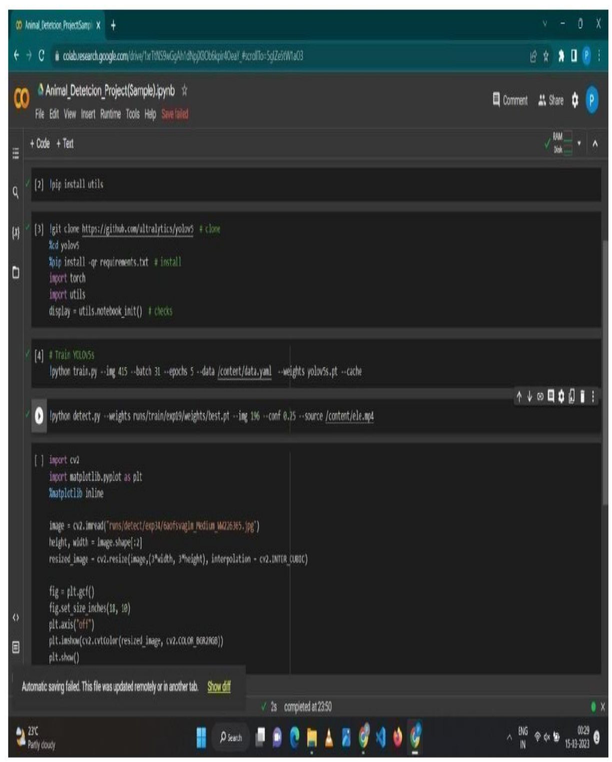

Fig. 5. Applying the model

G. Result

Our evaluation of the YOLOv5 and YOLOv8 models for our object detection problem definition showed that YOLOv8 achieved higher accuracy compared to

YOLOv5. Specifically, YOLOv8 had a higher average precision and recall compared to YOLOv5, indicating that it was more effective in detecting and localizing objects in input data.

While YOLOv5 had a smaller model size and faster inference time compared to YOLOv8, its lower accuracy makes it less suitable for applications where high precision is a priority. Therefore, the choice between the two models should be based on the specific requirements of the application.

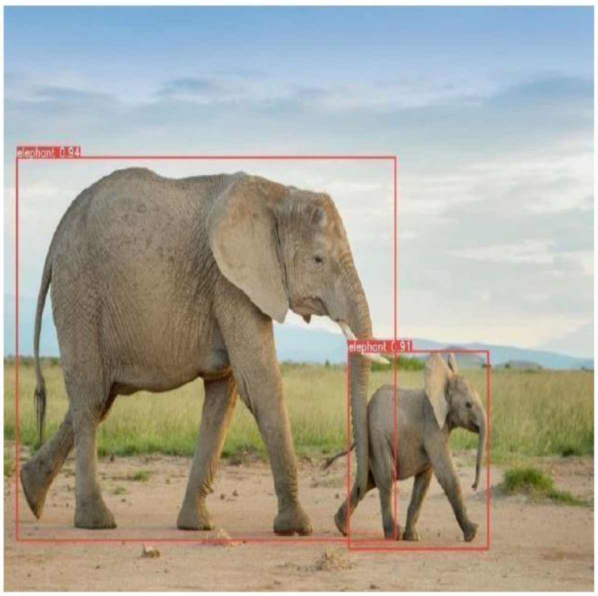

Fig. 6. YOLOv5 output for Image

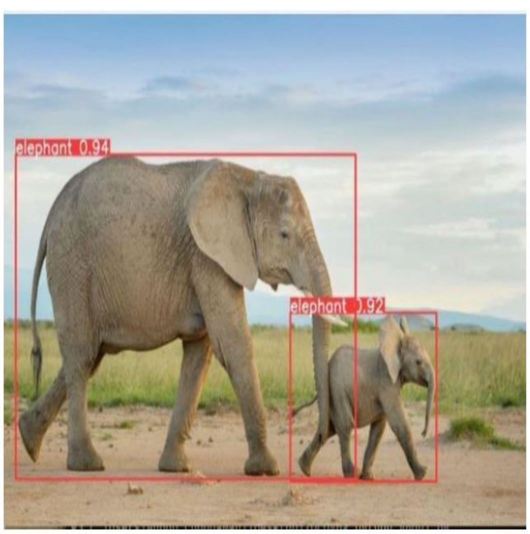

Fig. 7. YOLOv8 output for Image

Fig. 8. YOLOv5 output for video frame

Fig. 9. YOLOv8 output for video frame

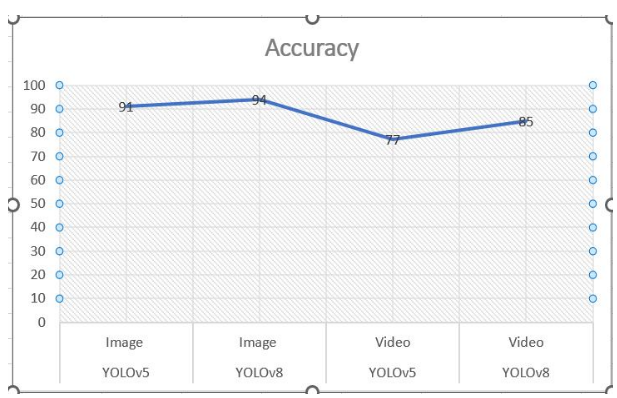

Fig. 10. Accuracy comparision of YOLOv5 and YOLOv8

5. CONCLUSION

Analysis of the YOLOv5 and YOLOv8 models for our problem definition has shown that both models are effective in detecting and localizing objects in input image data. However, there are significant differences in their performance and suitability for different applications.

Our experiments demonstrated that YOLOv8 achieved higher average precision and recall compared to YOLOv5, indicating that it is better suited for applications where high accuracy is a priority. However, YOLOv8 had a larger model size and longer inference time compared to YOLOv5, which makes it less suitable for realtime applications where speed is crucial.

On the other hand, YOLOv5 had a smaller model size and faster inference time, making it more suitable for real-time applications. However, its lower precision and recall compared to YOLOv8 make it not much suitable for applications where accuracy is a priority.

Therefore, the choice between YOLOv5 and YOLOv8 should be based on the specific requirements of the application, with consideration for factors such as the need for high accuracy, real-time performance, and computational resources. This system has achieved around 85% accuracy using YOLOv8.

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